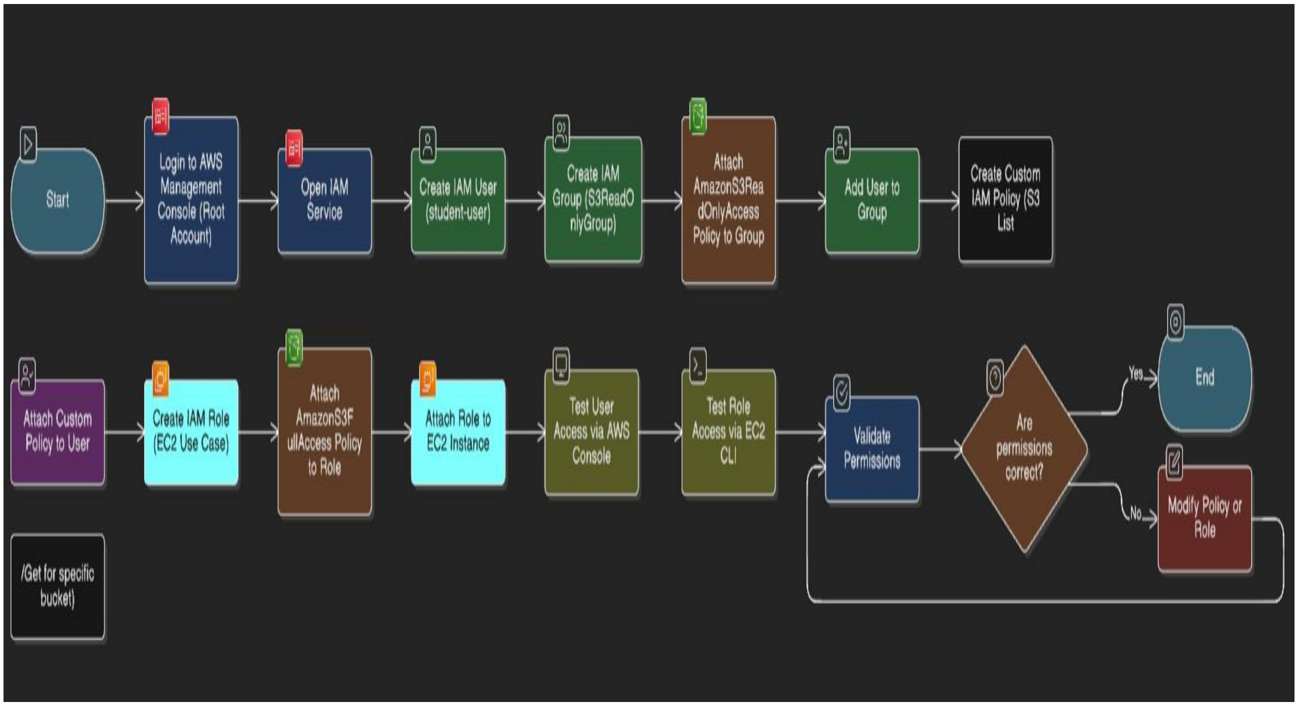
EXPERIMENT 5 : Identity and Access Management

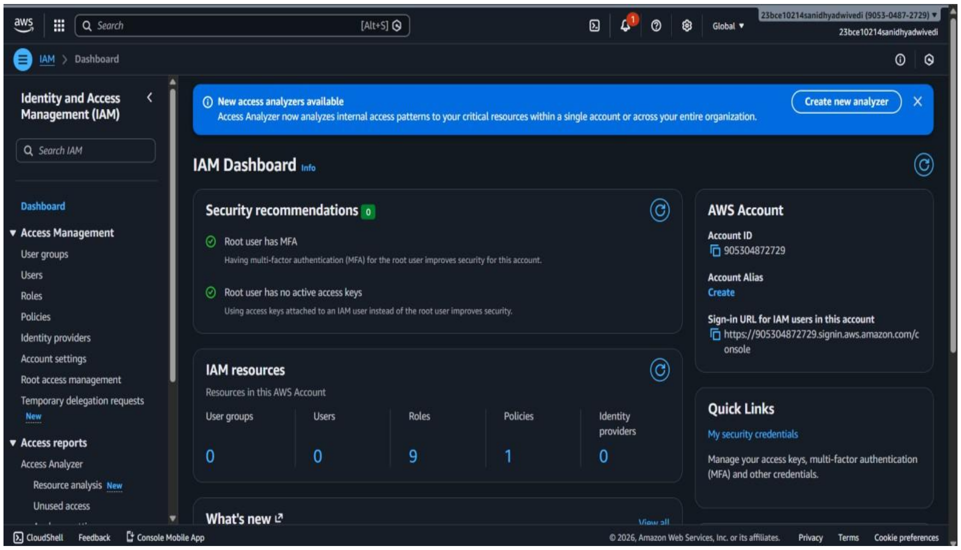


Step 1: Login to AWS Management Console

1. Go to: https://console.aws.amazon.com
2. Log in using your **root or IAM admin account**.

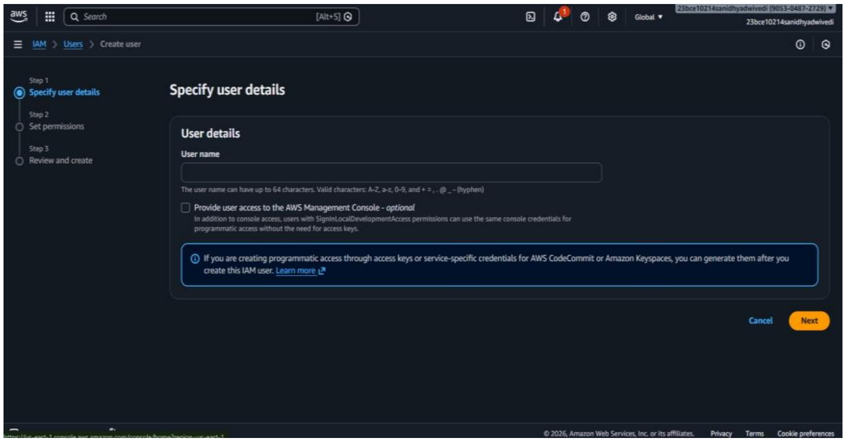
Step 2: Open IAM Service

1. In the AWS Console search bar → type **IAM**
2. Click **Identity and Access Management (IAM)**



Step 3: Create an IAM User

1. IAM Dashboard → Click **Users** → **Create user**
2. Enter:
    - **User name:** student-user
3. Select access type:
    - AWS Management Console access
    - Programmatic access (if CLI is needed)
4. Set Console password:
    - Choose **Autogenerated password** OR **Custom password**
    - Check *User must create a new password at next sign-in*
5. Click **Next**



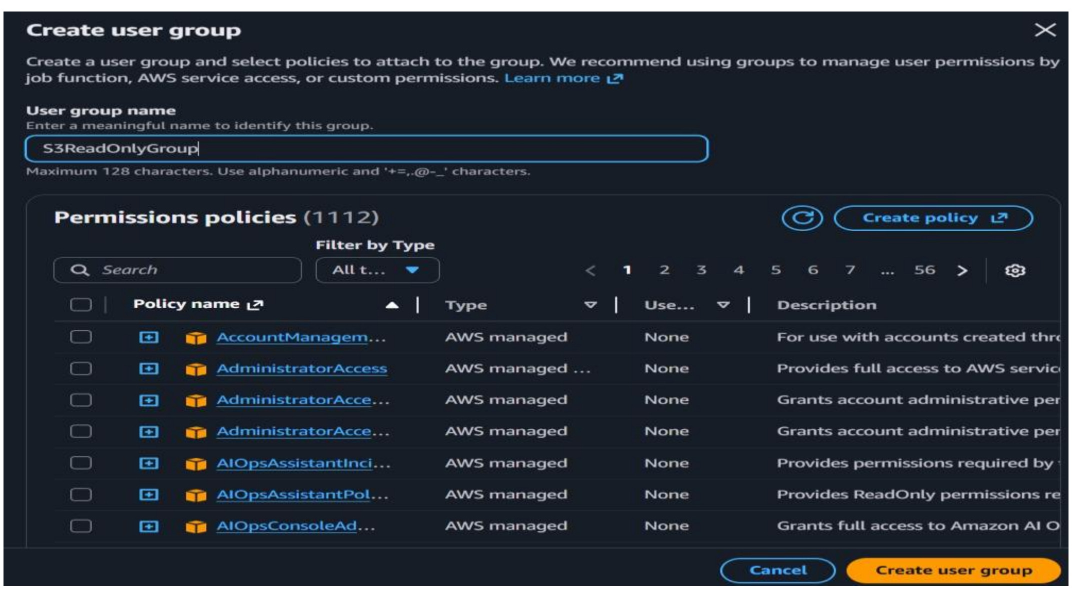
Step 4: Create a Group & Attach Policy

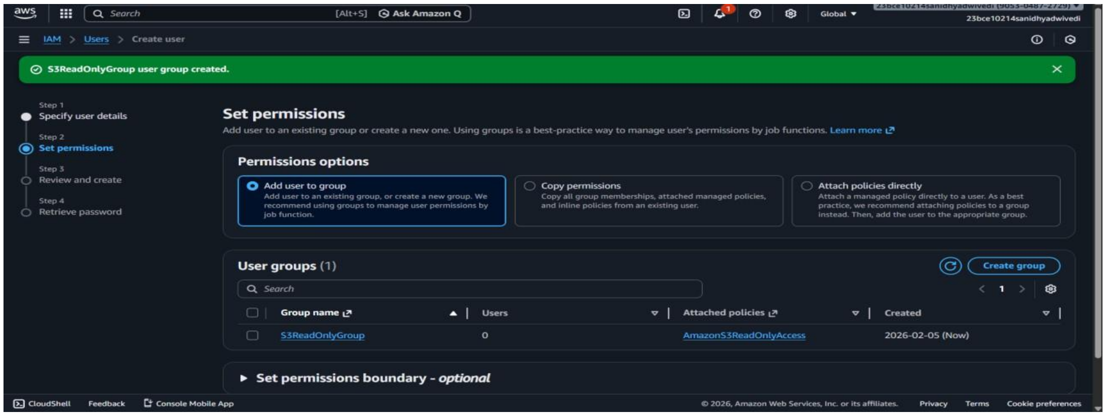
➤ Create Group

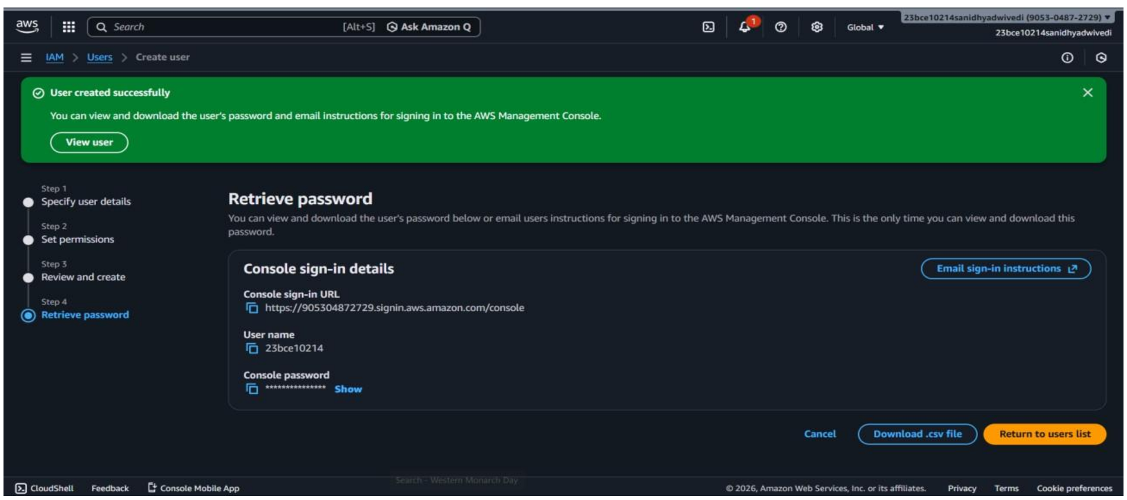
1. Click **Create group**
2. Group name: S3ReadOnlyGroup
3. Search and attach:
    - AmazonS3ReadOnlyAccess
4. Click **Create group**

➤ Add User to Group

1. Select S3ReadOnlyGroup
2. Click **Next**



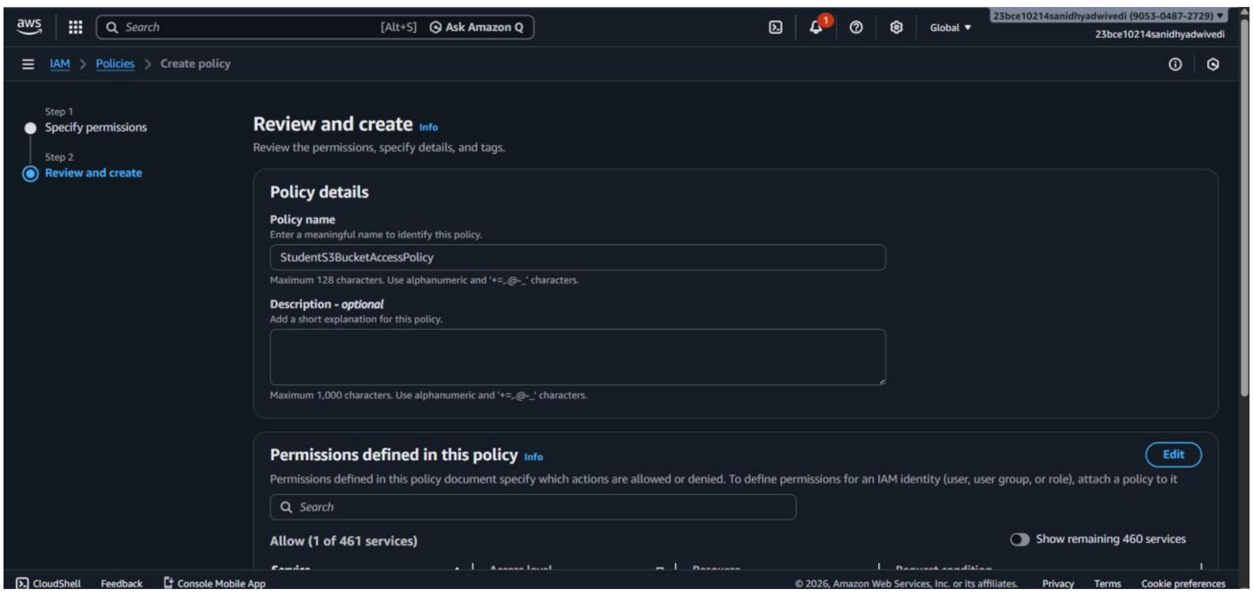




Step 5: Create a Custom Policy

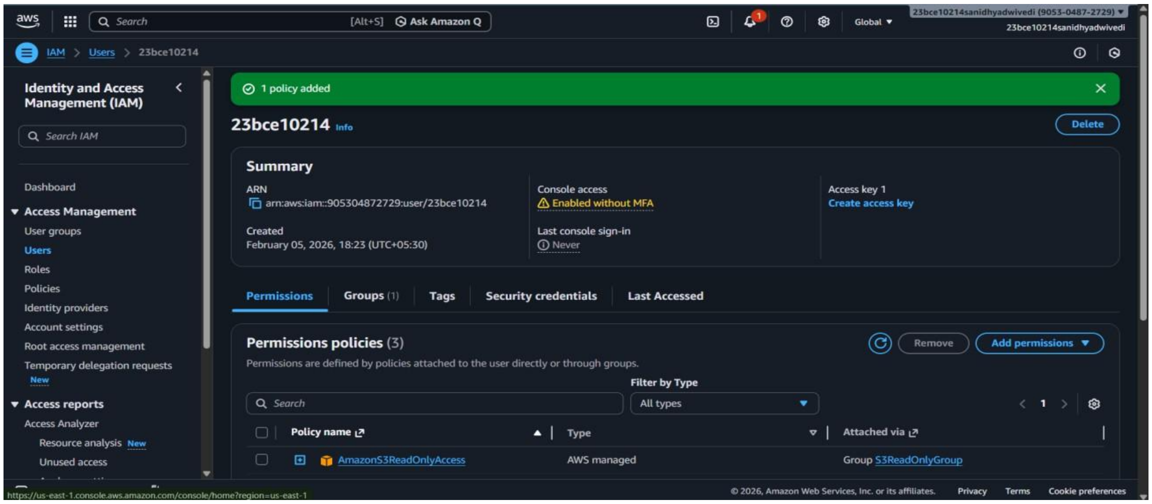
1. IAM → **Policies** → **Create policy**
2. Choose **JSON** tab
3. Now type this code:

```json
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:ListBucket",
        "s3:GetObject"
      ],
      "Resource": [
        "arn:aws:s3:::example-bucket",
        "arn:aws:s3:::example-bucket/*"
      ]
    }
  ]
}
```



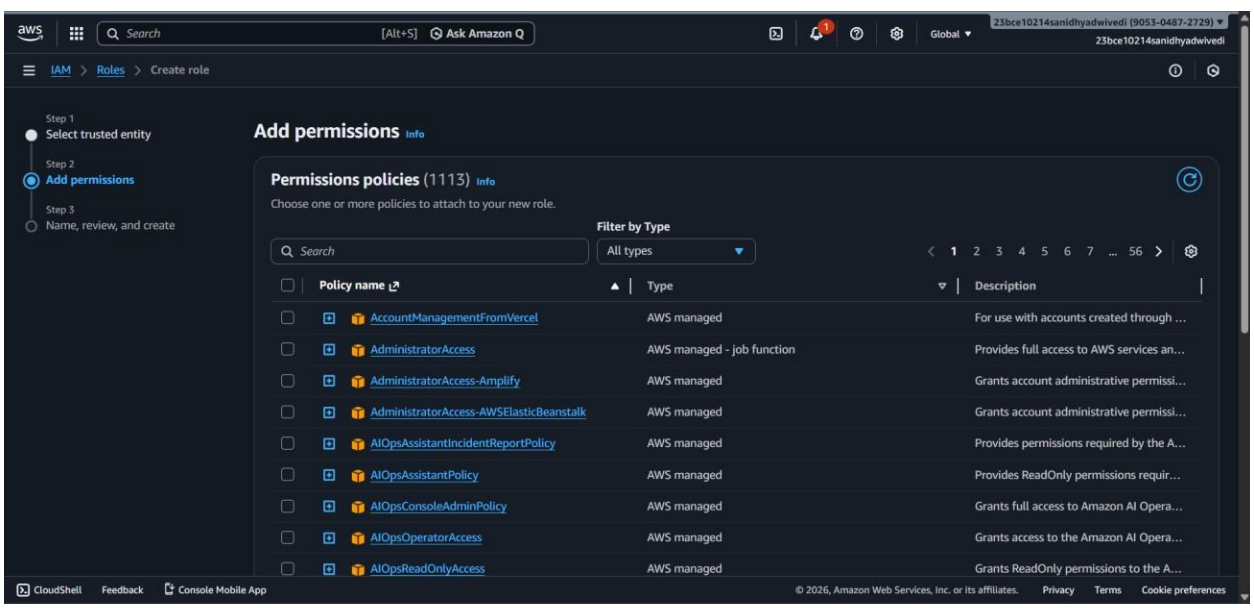
Step 6: Attach Custom Policy to User

1. IAM → **Users** → Click student-user
2. Go to **Permissions** tab → **Add permissions**
3. Choose **Attach policies directly**
4. Search → StudentS3BucketAccessPolicy
5. Select → Click **Next** → **Add permissions**



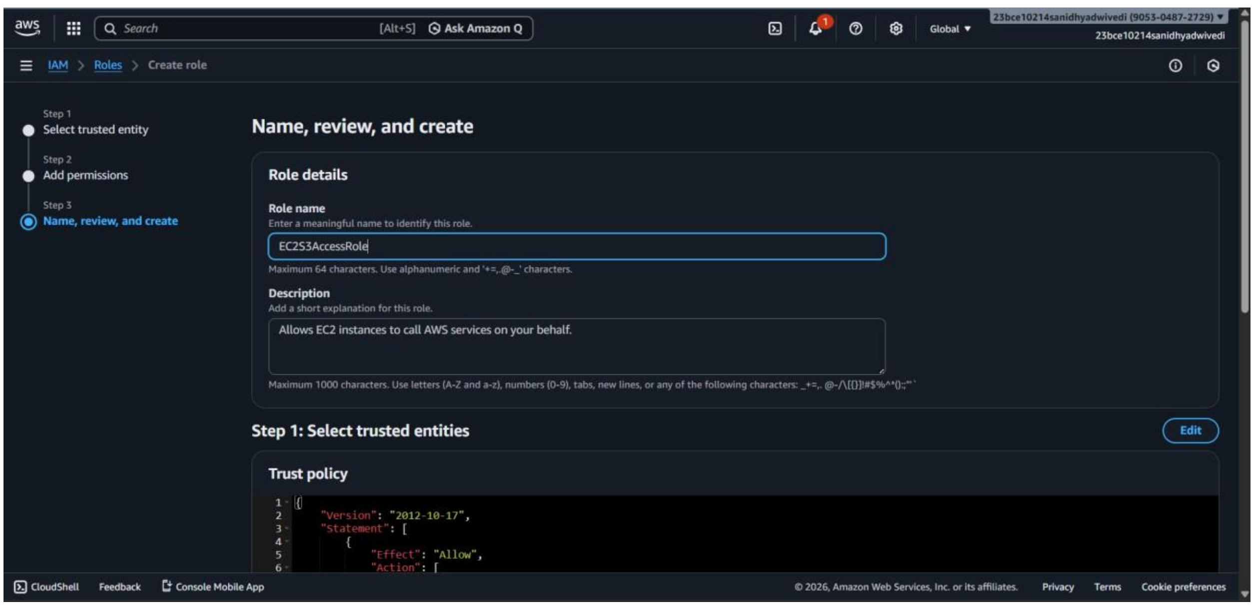
Step 7: Create a Role

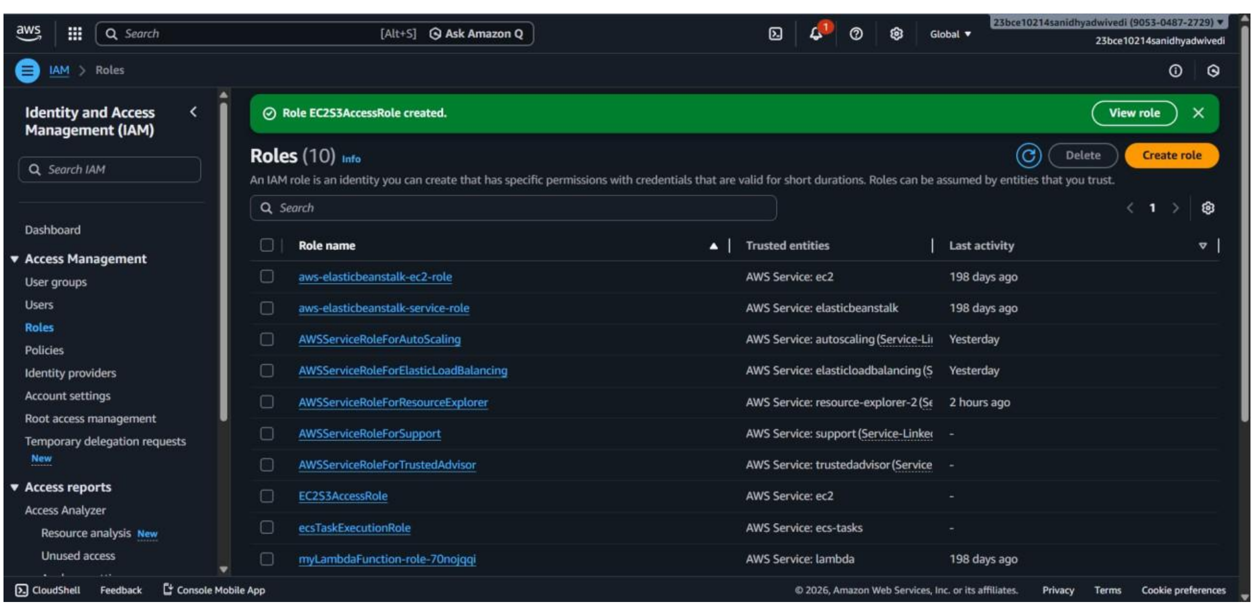
1. IAM → **Roles** → **Create role**
2. Select trusted entity:
   - AWS service
3. Use case:
   - EC2
4. Click **Next**



Step 8: Attach Policy to Role

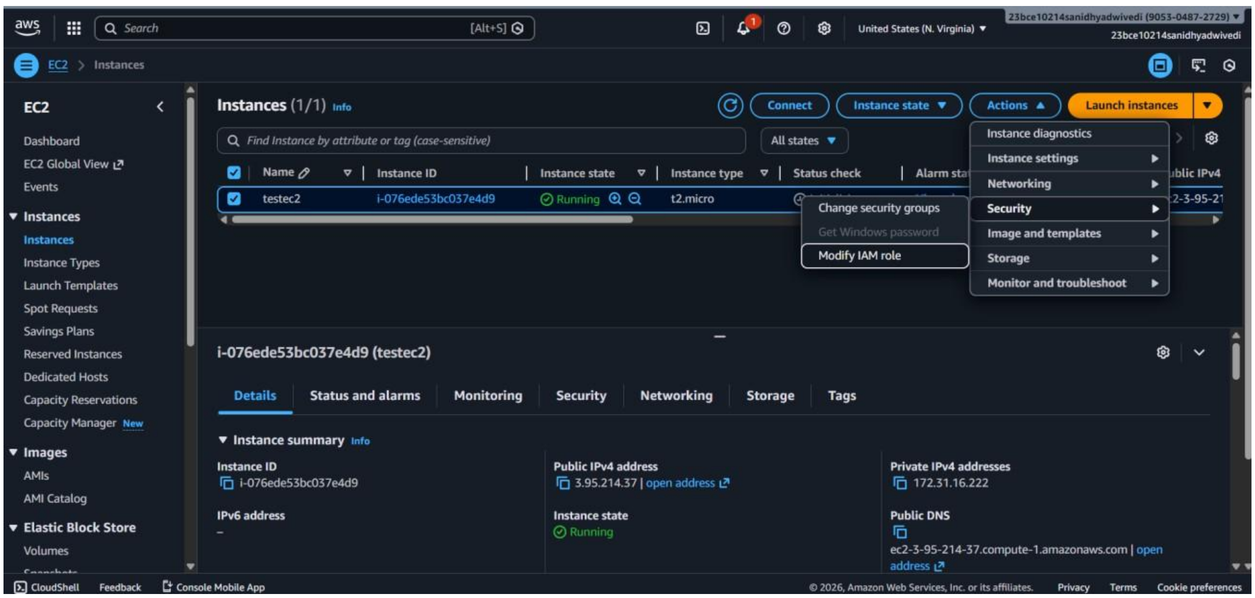
1. Search and select:
   - AmazonS3FullAccess
2. Click **Next**
3. Role name: EC2S3AccessRole
4. Click **Create role**

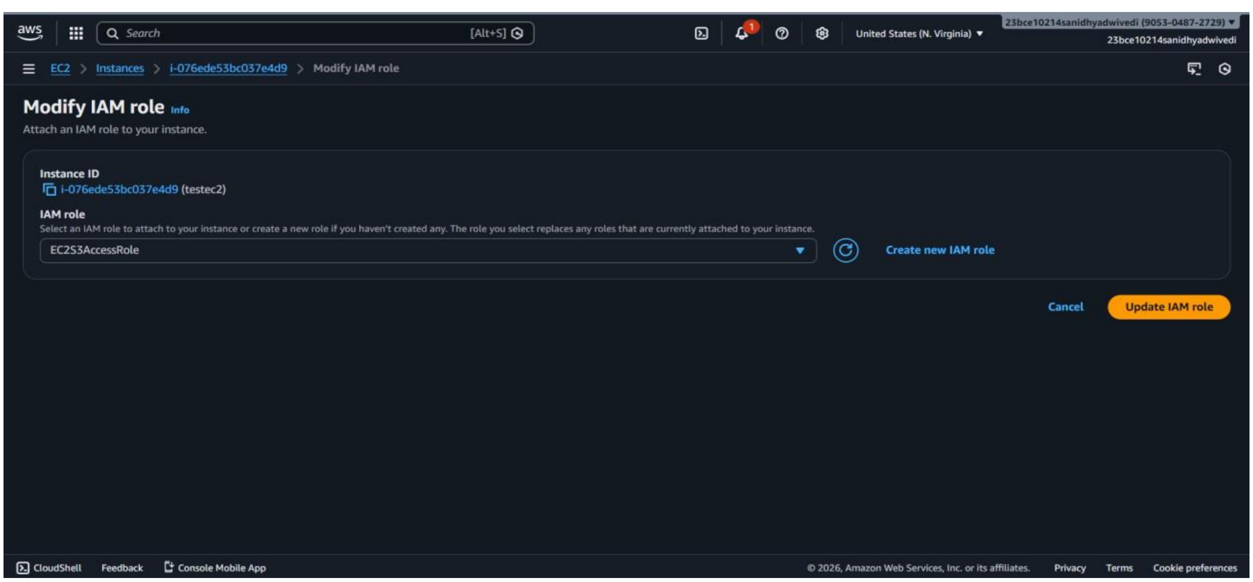


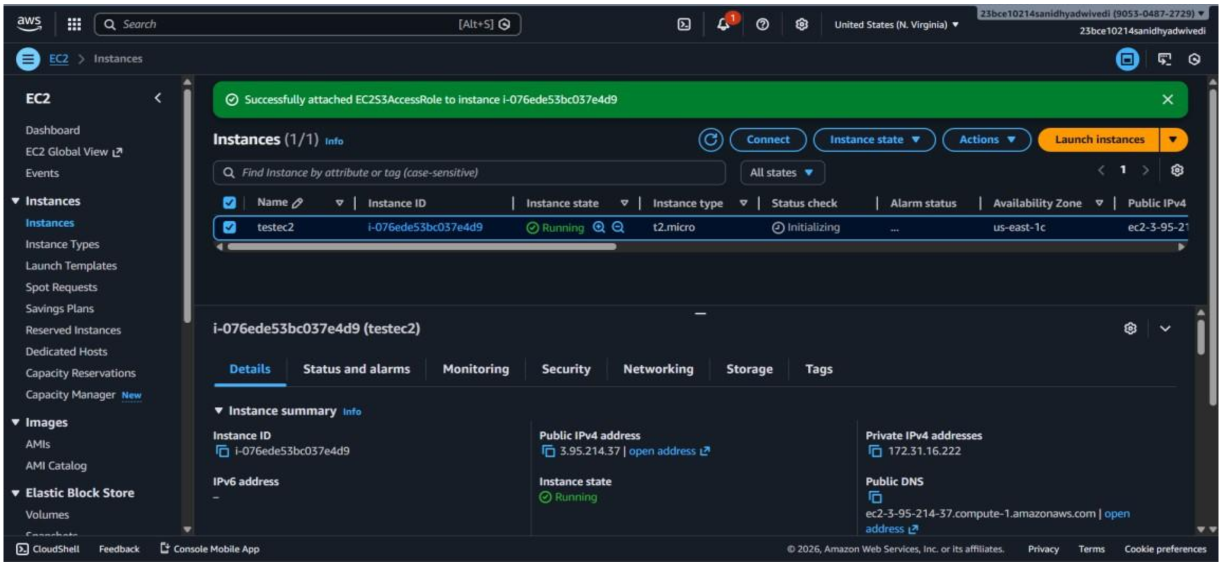


Step 9: Attach Role to EC2 Instance

1. Go to **EC2 Dashboard**
2. Instances → Select your EC2 instance
3. Actions → Security → Modify IAM role
4. Select role: EC2S3AccessRole
5. Click **Update IAM role**







Step 10: Test Permissions

➤ Test User Access

1. Log out → Login using **student-user**
2. Try:
   - Open **S3**
   - You should be able to:
     - List bucket
     - Download objects
   - \+ Not upload/delete (if read-only)